

Tutorial: Pandas

Go to the Moodle data Science directory and get the file entitled "120-years-of-olympic-history-athletes-and-results.csv". It contains information on the results of all athletes who participated in the Olympic Games from 1896 to 2016.

1. Read the CSV file
2. Describe the database: number of rows, columns.
3. Header management: remove the English header and replace it with another language header.
4. Go back to the English language. Set the first column of the file as the index of the table.
5. Select only columns containing athletes' names, weights and heights using column positions. Answer the same question by using the column names.
6. Explore the data types of the dataframe
7. If columns contain dates, use the "parser" method to format those dates correctly.
8. Read the Excel file
9. Read the database
10. Read the JSON dataset
11. Select the column that corresponds to the weight of the athletes in the table "120 years ... " and display it.
12. Ask Pandas to select columns 1 and 5, respectively Name and Weight, with the usecols option.
Also specify to Pandas that you want that the created object to be of Series class thanks to the squeeze option.
Set the first column of this object to match the indexes, with the index_col option..
13. Get the weights of athletes "Antti Sami Aalto" and "Andrzej ya".
14. Create a series, which contains four values and whose index names are 1, 3, 2, and 0, respectively.
15. Select only athletes with a weight strictly greater than 90 kg.
16. Same question as above with weights strictly above 90 and strictly below 100.
17. How to view the following table:

```

1000    84.0
1001    84.0
1002    84.0
1003    73.0
1004    54.0
...
271111    89.0
271112    59.0
271113    59.0
271114    96.0
271115    96.0
Name: Weight, Length: 270116, dtype: float64

```

18. How to view the following table:

```

0    80.0
1    60.0
2    NaN
3    NaN
4    82.0
5    82.0
6    82.0
7    82.0
8    82.0
9    82.0
10   75.0
11   75.0
12   75.0
13   75.0
14   75.0
15   75.0
16   75.0
17   75.0
18   72.0
19   72.0
Name: Weight, dtype: float64

```

19. How to determine the size of our series containing the weight of athletes.
20. Get different statistics from the series giving the weights of the athletes.
21. Index the dataframe by choosing the names column as the index, and display Usain Bolt's data.
22. Compare Usain Bolt's results with those of Justin Alexander Gatlin
23. Select all the women who have participated in the Olympics and won a gold medal.
24. Give the number of missing values per variable.
25. Remove athletes whose height is not listed in the dataset.
26. Replace the missing values in the Age, Height, and Weight columns with the average value in each column, respectively.
27. Detect duplicate rows. How many times are they duplicated?
28. Delete duplicate rows. How many lines are left?
29. Display the shape of the resulting dataframe after deleting the duplicate rows.
30. Execute:

```
data.groupby("Sport").mean().loc[:,['Age','Height','Weight']]
```

Explain the dataframe obtained.

31. How to get the average age, height and weight by sport and gender of athletes.